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Studierichting: Informatica
Oefeningenreeks 1C nr. 48.1

Opgave: $z^6 = 1$

$r = \sqrt{1^2 + 0^2} = \sqrt{1} = 1$

$\tan\theta = \dfrac{0}{1} = 0 \Rightarrow \theta = \mathrm{Bgtan}\,0 = 0$

$\Rightarrow w = \sqrt[6]{1}\,\mathrm{cis}\left(\dfrac{0 + k \cdot 2 \cdot \pi}{6}\right)$

$w_0 = 1 \cdot \mathrm{cis}\,(0) = 1$

$w_1 = 1 \cdot \mathrm{cis}\left(\dfrac{2\pi}{6}\right) = \mathrm{cis}\left(\dfrac{\pi}{3}\right) = \dfrac{1}{2} + \dfrac{\sqrt{3}}{2}i$

$w_2 = 1 \cdot \mathrm{cis}\left(\dfrac{4\pi}{6}\right) = \mathrm{cis}\left(\dfrac{2\pi}{3}\right) = -\dfrac{1}{2} + \dfrac{\sqrt{3}}{2}i$

$w_3 = 1 \cdot \mathrm{cis}\,(\pi) = -1$

$w_4 = 1 \cdot \mathrm{cis}\left(\dfrac{8\pi}{6}\right) = \mathrm{cis}\left(\dfrac{4\pi}{3}\right) = -\dfrac{1}{2} - \dfrac{\sqrt{3}}{2}i$

$w_5 = 1 \cdot \mathrm{cis}\left(\dfrac{10\pi}{6}\right) = \mathrm{cis}\left(\dfrac{5\pi}{3}\right) = \dfrac{1}{2} - \dfrac{\sqrt{3}}{2}i$

References